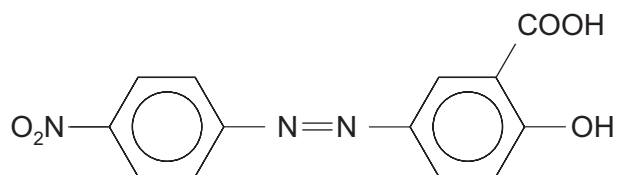


10. (a) The azo dye Mordant Orange 1 has the formula shown below.



It is used as a pH indicator for acid-base titrations. At a pH of 10.1 the dye is yellow but at a pH of 12.1 it is orange-red.

- (i) Give the displayed formulae of the two starting compounds that react, via a diazonium compound, to give Mordant Orange 1. [2]

- (ii) The table shows the colour seen and the colour absorbed for a solution of Mordant Orange 1 as the pH increases.

Colour seen	yellow	orange-red
Colour absorbed	violet	greenish-blue
pH increasing $\longrightarrow$		

State how the wavelength of the light **absorbed** changes with pH, giving your reasoning. [1]

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- (iii) The visible spectrum of the dye at pH 10.1 shows that it has a maximum absorption at 385 nm.

Calculate the frequency of the radiation being absorbed at 385 nm. [2]

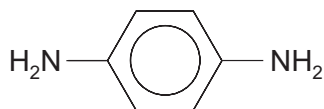
Frequency = Hz

- (iv) Use your answer to (iii) to calculate the energy of the maximum absorption at 385 nm, giving your answer in kJ mol^{-1} . [3]

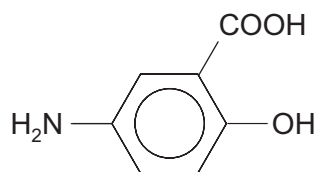
Energy = kJ mol^{-1}



- (b) Mordant Orange 1 can be reduced by the enzymes from certain bacteria to produce 1,4-diaminobenzene and 5-amino-2-hydroxybenzoic acid



1,4-diaminobenzene



5-amino-2-hydroxybenzoic acid

- (i) 1,4-diaminobenzene can also be made from 4-nitrophenylamine.

State the reagent(s) necessary for this reaction.

[1]

- (ii) Use the displayed formula of 1,4-diaminobenzene to help you describe its ^1H high resolution NMR spectrum. Explain your reasoning.
Reference to the position of the signals is not required.

[3]

- (iii) Give the displayed formula of the organic compound formed when 5-amino-2-hydroxybenzoic acid reacts with aqueous sodium hydroxide in a 1:1 molar ratio.

[1]



- (iv) 5-Amino-2-hydroxybenzoic acid can be used in medicine to treat a number of conditions. It is believed to work by removing free radicals. One of these radicals is the hydroxyl radical. This can be produced by the homolytic fission of hydrogen peroxide, H_2O_2 , where the $\text{O}-\text{O}$ bond is broken.

Draw a dot and cross diagram of the hydroxyl radical, showing the outer electrons for each atom. [1]

Examiner
only

14



10. (a) Butylamine can be made from propan-1-ol in a three-stage process.

Give the reagent used and the structural formula of the product for each stage.

[3]

Stage	Reagent used	Structural formula of product
1		
2		
3		$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$

(b) Below 10°C , phenylamine reacts with nitric(III) acid, HNO_2 , to give a benzenediazonium compound, which can react with phenol to give an azo dye.

(i) State how nitric(III) acid is prepared for use in this reaction.

[1]

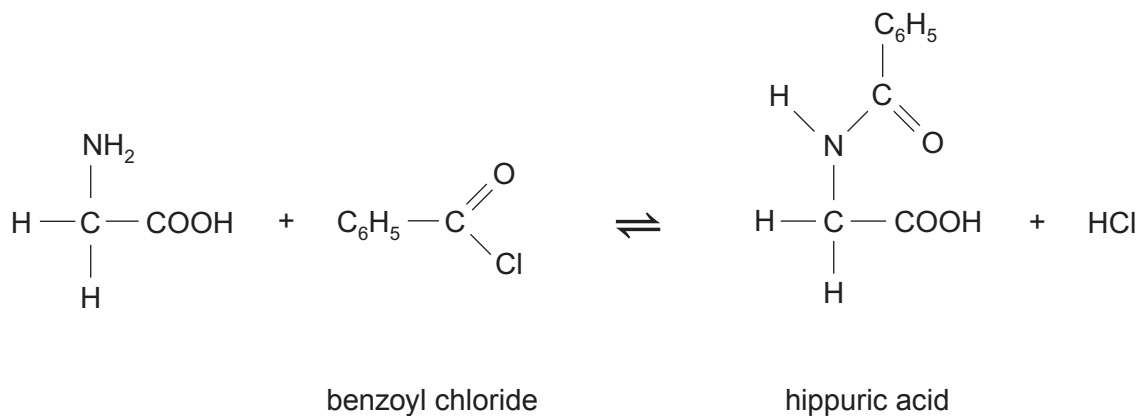
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(ii) Give the equation for the reaction of benzenediazonium chloride with phenol to produce an azo dye.

[1]



- (c) Hippuric acid, so called because it is found in the urine of horses, can be made by reacting aminoethanoic acid and benzoyl chloride.



In an experiment the percentage yield was 90 %.

Benzoyl chloride is only slowly hydrolysed by alkalis. A teacher suggested the following idea for increasing the percentage yield.

‘The experiment could be carried out in the presence of aqueous sodium hydroxide.’

Comment on the validity of this suggestion and any effect it may have on the yield.
Explain your answer.

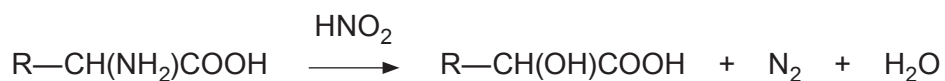
[1]

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- (d) A straight chain α -amino acid of general formula $R-CH(NH_2)COOH$ reacts with nitric(III) acid to give nitrogen gas.



In an experiment 7.87 g of the amino acid was dissolved in water and the solution made up to 500 cm^3 . A 25.0 cm^3 sample of this solution was reacted with an excess of nitric(III) acid and gave 73.5 cm^3 of nitrogen measured at 298 K and 1 atm pressure.

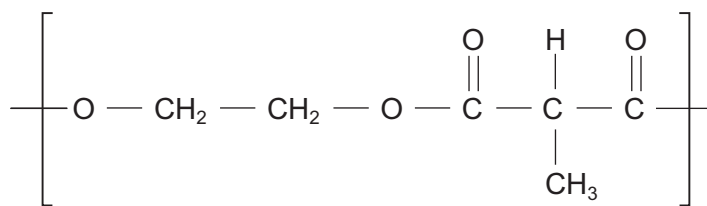
Calculate the molar mass of the amino acid and hence its structural formula. [4]

Molar mass =

Structural formula



- (e) Polyesters can be made from a dicarboxylic acid and a diol, such as ethane-1,2-diol.

polyester **E**

- (i) State the name of the dicarboxylic acid that reacts with ethane-1,2-diol to give polyester **E**. [1]

- (ii) Complete the table to describe the high resolution ^1H NMR spectrum of polyester **E**. [3]

Protons	Splitting pattern	Relative peak area
$-\text{CH}_2-\text{CH}_2-$		
$-\text{CH}-$		
$-\text{CH}_3$		

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2

**GCE A LEVEL**

1410U40-1



S19-1410U40-1

TUESDAY, 11 JUNE 2019 – AFTERNOON

CHEMISTRY – A2 unit 4
Organic Chemistry and Analysis

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
Section A 1. to 7.	10	
Section B 8.	13	
9.	15	
10.	13	
11.	13	
12.	16	
Total	80	

1410U401
01**ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11(a)(i)**.

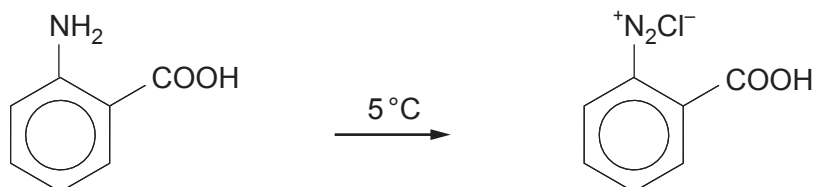
If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



JUN191410U40101

10. (a) The acid-base indicator methyl red can be made from 2-aminobenzenecarboxylic acid.

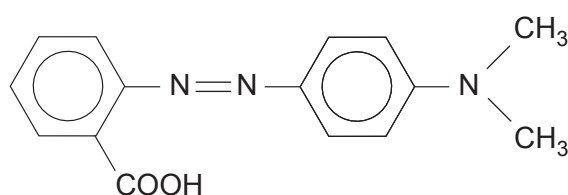
- (i) In the first stage of the reaction, a benzenediazonium compound is made from this acid.



2-aminobenzenecarboxylic acid

State the reagent(s) necessary to produce this diazonium compound from the acid. [1]

- (ii) The benzenediazonium compound is then coupled with an amine to make methyl red.



methyl red

Draw the structure of the amine used. [1]

- (iii) The visible spectrum of this azo dye at a certain pH shows an absorption maximum at 410 nm.

Calculate the frequency that corresponds to this wavelength. [2]

Frequency = Hz



- (b) Some azo dyes are permitted for use as food colours but their concentration is strictly controlled. The concentration present can be found by colorimetry. The absorption of a series of standard solutions is measured and a straight line calibration graph is drawn.

- (i) The molar mass of an azo dye is 272 g mol^{-1} .

Calculate the mass of the azo dye needed to prepare 250 cm^3 of a standard solution of concentration $0.0075 \text{ mol dm}^{-3}$. [2]

Mass required = g

- (ii) Absorption (A) and concentration (c) are related by the equation

$$A = k c \quad \text{where } k \text{ is a constant}$$

The calibration graph shows that an absorption of 1.44 is measured for a solution of concentration $0.0096 \text{ mol dm}^{-3}$.

Calculate the concentration of a solution that gave an absorption of 1.03. [2]

Concentration = mol dm^{-3}

- (c) (i) When ethanoic acid is heated with urea, NH_2CONH_2 , the products are ethanamide, carbon dioxide and ammonia.

Give the equation for this reaction. [1]

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- (ii) Suggest why an amide can act as a base. [1]

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- (iii) When ethanamide is heated with a dehydrating agent the organic product is ethanenitrile.

Use the **Data Booklet** to state and explain how characteristic infrared absorption values change during the reaction. In your answer you should relate the relevant bonds to their corresponding absorption frequencies. [2]

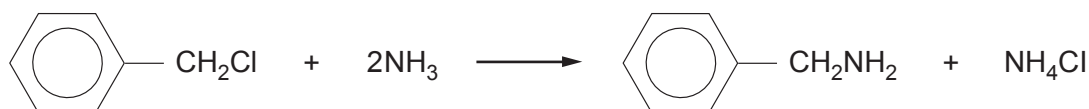
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- (iv) Some amines can be made by the reaction of a chlorine-containing compound and ammonia. For example (chloromethyl)benzene reacts with ammonia to give phenylmethanamine.



Suggest why phenylamine cannot easily be made by this method from chlorobenzene and ammonia. [1]

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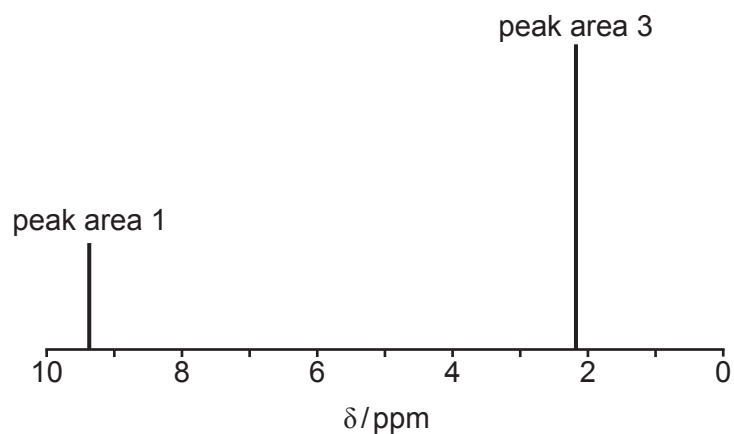
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10. (a) Compound **G** contains only carbon, hydrogen and oxygen.

It has a molar mass of 72 g mol^{-1} of which 50.0% is carbon. The compound reacts positively with Tollens' reagent and gives a yellow solid when treated with alkaline iodine solution. It reacts with sodium tetrahydridoborate(III) to give a new compound which has a molar mass of 76 g mol^{-1} .

The high resolution ^1H NMR spectrum of compound **G** is shown below.



Use **all** of this information to deduce a structure for compound **G**.

You should comment on how each piece of data has helped you to deduce the structure.
[6 QER]

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- (b) The absorption spectrum of compound **G** shows a maximum absorption at 450 nm.
Light energy measured in kJ mol^{-1} (E) is related to wavelength (λ) by the equation

$$E = \frac{\text{constant}}{\lambda}$$

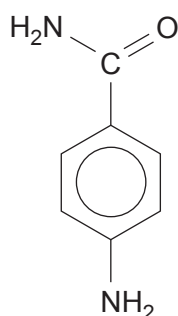
The energy of light of wavelength 656 nm is 183 kJ mol^{-1} .

Calculate the energy of the maximum absorption of compound **G** at 450 nm. [2]

Energy = kJ mol^{-1}



(c) The formula of 4-aminobenzamide is



(i) When this compound is warmed with excess aqueous sodium hydroxide, only the amide group reacts.

I. Give the structure of the organic product. [1]

II. State the nature of the attacking reagent and explain why the NH_2 group bonded to the benzene ring is not removed. [2]

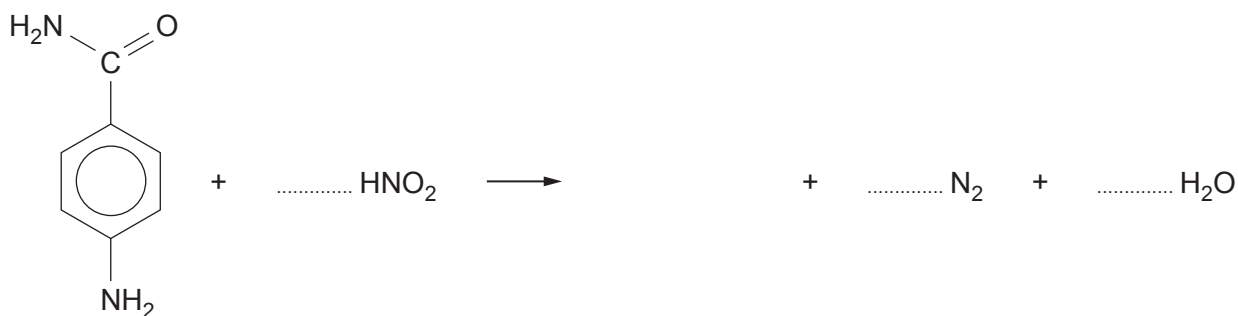
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(ii) Both the functional groups in 4-aminobenzamide are attacked by nitric(III) acid at room temperature, giving 4-hydroxybenzoic acid, nitrogen gas and water as the products.

Complete the equation for this reaction. [2]



10. (a) Benzene reacts with bromine in the presence of a catalyst.
Give the mechanism for this reaction and explain how the Br — Br bond becomes polarised during the reaction.
Suggest why, in the absence of this catalyst, there is very little reaction between benzene and bromine under normal conditions. [6 QER]

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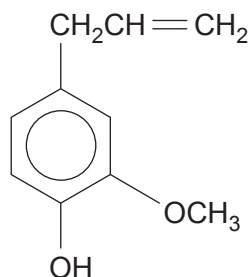
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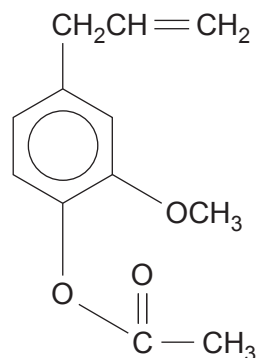
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- (b) Eugenol is the main constituent of clove oil, together with smaller quantities of eugenyl ethanoate.



Eugenol



Eugenyl ethanoate

- (i) Describe what is seen if a few drops of iron(III) chloride solution are added to a solution of eugenol. [1]

- (ii) State the colour change that occurs if a few drops of aqueous bromine are added to a solution of eugenol. [1]

- (iii) If an excess of bromine is added to eugenol, a new compound is formed that contains the following percentages by mass of each element.

C 24.9% H 2.1% O 6.6%, the remainder being the % of bromine

Use this information to calculate the empirical and molecular formulae of this brominated compound and suggest a possible structure for it. [5]



- (iv) Explain how adding aqueous sodium hydroxide at room temperature to a solution of eugenol and eugenyl ethanoate dissolved in trichloromethane, enables the two compounds to be separated. [2]
Trichloromethane and water are immiscible.

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